

Marchmello01: Topology-Safe Adaptive Lesion Fusion for Interactive PET/CT Segmentation

Marchmello01

Independent researcher

Abstract. We describe an interactive whole-body tumor segmentation system for FDG and PSMA PET/CT. A five-fold nnU-Net residual-encoder ensemble produces the initial mask, followed by tracer-specific calibration and a prompt-conditioned distance-transform model. We introduce conservative interaction changes: clicks are mapped to cropped model space using one crop-origin subtraction; donor components are validated independently to prevent a newly detected lesion from hiding a harmful bridge; PSMA foreground corrections are enabled up to a fixed 128-component burden; and explicitly annotated PSMA background voxels are removed only when removal cannot fragment a predicted component. Paired replay on local interactive fixtures showed no observed FDG regression and improved one PSMA trajectory from 0.8278 to 0.8343 AUC-Dice and from 0.9296 to 0.9502 AUC-DMM/F1. The method is stateless, deterministic with respect to cumulative scribbles, and is released with complete upstream attribution.

Keywords: autoPET V · interactive segmentation · PET/CT · topology

1 Introduction

autoPET V evaluates lesion segmentation across FDG and PSMA PET/CT under iterative simulated and clinician-driven scribbles. High voxel overlap alone is insufficient: a useful method must also create missed lesion instances and remove false-positive instances without merging anatomically separate tumors. Our submission targets this topological failure mode while preserving strong initial segmentation from public AutoPET-III and AutoPET-IV components.

2 Methods

2.1 Data

Training data The pretrained components use the public data and preprocessing described by the upstream AutoPET-III LesionTracer and AutoPET-IV LesionLocator releases. Our changes introduce no additional training data or weight optimization.

Validation data Local validation used four FDG and four PSMA interactive fixtures, with cumulative challenge-format scribbles replayed for five steps. These fixtures were used for paired engineering comparisons and are not presented as an unbiased estimate of hidden-test rank.

2.2 Data preprocessing

CT and PET are resampled and normalized by the frozen nnU-Net plans. Predictions are returned to the CT reference grid, and the output copies its size, spacing, origin, and direction. Input-grid click coordinates are mapped into cropped model space by subtracting the crop origin once and then applying the crop-to-resampled scale. No metadata identifying the tracer is required.

2.3 Initial segmentation and tracer routing

Three frozen classifiers route each scan to FDG or PSMA. The initial mask is a five-fold AutoPET-III LesionTracer ResEncL ensemble without mirror test-time augmentation. FDG uses a probability threshold of 0.47 and burden-adaptive small-component filtering. PSMA uses threshold 0.50, five-voxel filtering, and the upstream frozen logistic component pruner at threshold 0.86.

2.4 Algorithm and topology-safe interaction

Later calls reconstruct the initial mask and run the fold-0 EDT LesionLocator with all cumulative scribbles. Each clicked foreground donor component is considered independently. It is accepted only if union with the current mask does not reduce the connected-component count, preventing lesion bridges. For PSMA, foreground correction requires at least six cumulative foreground points and at most 128 initial components. Each connected background stroke is also handled independently: only its explicitly marked voxels are erased, and the erase is accepted only when it does not increase component count. FDG background behavior remains unchanged from the upstream system.

2.5 Data postprocessing

FDG predictions use a probability threshold of 0.47 and burden-adaptive connected-component filtering. PSMA uses threshold 0.50, five-voxel filtering, and a frozen logistic component pruner with false-positive threshold 0.86. All interactive unions and erasures are checked using 18-connected components.

2.6 Training and test parameters

All network weights are unchanged pretrained weights. Consequently, no new optimizer, schedule, augmentation, or training hardware is applicable to this derivative. Inference uses five initial-model folds, one interactive fold, disabled mirror TTA, binary connected components, and cumulative stateless scribble replay. The submitted image targets an NVIDIA T4 with 16 GB VRAM.

2.7 GitHub repository

Code, tests, build recipes, notices, and checkpoint integrity hashes are released under a permissive license at github.com/Marchmello01/autopet-v-topology-safe-adaptive.

3 Results

On paired local replay, the topology gate was identical to upstream on ordinary trajectories. Safe background removal raised PSMA1 AUC-Dice from 0.8006 to 0.8011 with unchanged F1. Relaxed, topology-checked correction raised PSMA2 AUC-Dice from 0.8278 to 0.8343 and AUC-DMM/F1 from 0.9296 to 0.9502. PSMA4 was unchanged (0.8480 Dice, 0.8611 F1), and the conservative path was retained for the highly fragmented 178-component PSMA1 scan. These local fixtures support selection of the submitted policy but are not a substitute for hidden-test evaluation. The frozen container subsequently obtained preliminary-test Dice 0.854649, F1/DMM 0.834179, and mean position 3.5. The preliminary set contains only five cases and is reported as an implementation check, not as a final-test performance claim.

4 Discussion and conclusion

The method changes interaction logic rather than claiming newly trained model weights. Its main advantage is a simple invariant: one useful new lesion may not compensate for a separate harmful bridge in the same update. The 128-component ceiling is deliberately conservative and should be re-estimated on a larger independent validation set. This topology-safe derivative improved the targeted PSMA case without observed regression in the evaluated fixtures.

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Disclosure of Interests The author declares no competing interests.

References

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Table 1. Algorithm details.

Team	Algorithm	Preprocessing	Postprocessing	Augmentation
Marchmello01	Tracer Adaptive LesionLocator Ensemble	nnU-Net PET/CT normalization and resampling	Tracer calibration; topology-safe component fusion	Upstream pretrained models; no new training
TTA	Ensembling	Framework	Architecture	Training loss
None	Five-fold K0; one interactive fold; three-model tracer vote	nnU-Net v2 / PyTorch	3-D residual- encoder U-Nets	Unchanged upstream Dice and cross-entropy objectives
Training data	Dimensionality	Pretrained models	Hardware	
Public AutoPET FDG/PSMA cohorts; no new training	3-D whole-body PET/CT with sliding-window inference	Public AutoPET-III initializer and AutoPET-IV interaction model	NVIDIA T4 16 GB inference target	